

# Sebeni morphotactic post-training on MULTI13: SFT, DPO and APO arms with SmolLM2-135M

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## Abstract

Three post-training arms of Sebeni, a morphotactic generation toolkit for extremely low-resource Manding languages, on the MULTI13 scope (13 languages). Each arm distils the grammar and dictionary, fine-tunes SmolLM2-135M with 4-bit LoRA for 500 steps on the packaged 300-sample dataset with a 10% held-out split, and scores the held-out test set (phi, MER, MCS, UWEC). Code: [github.com/mlsftwrs/sebeni](https://github.com/mlsftwrs/sebeni) at commit 8e66775, with K-Veritas capture added to the CLI.

### About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], [3], [4], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00001v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

## 1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

| Report | Label   | Sealed     | Runs | Invocations | Signer                         |
|--------|---------|------------|------|-------------|--------------------------------|
| r1 [2] | SFT arm | 2026-10-05 | 4    | 4           | Signed by the K-Veritas server |
| r2 [3] | DPO arm | 2026-10-05 | 4    | 4           | Signed by the K-Veritas server |
| r3 [4] | APO arm | 2026-10-05 | 4    | 4           | Signed by the K-Veritas server |

## 2 Results

### r1: SFT arm

| Run | Metric                   | Final value | Step |
|-----|--------------------------|-------------|------|
| 3   | loss                     | 1.30145     | 500  |
| 3   | eval_loss                | 1.32598     | 500  |
| 3   | eval_runtime             | 22.1618     | 500  |
| 3   | eval_samples_per_second  | 16.244      | 500  |
| 3   | eval_steps_per_second    | 2.031       | 500  |
| 3   | eval_entropy             | 1.36176     | 500  |
| 3   | eval_num_tokens          | 1.52793e+07 | 500  |
| 3   | eval_mean_token_accuracy | 0.768744    | 500  |
| 3   | epoch                    | 2.4642      | 500  |
| 4   | phi_bam                  | 0.823658    | 0    |
| 4   | mer_bam                  | 0           | 0    |
| 4   | mcs_bam                  | 0           | 0    |

| Run | Metric   | Final value | Step |
|-----|----------|-------------|------|
| 4   | uwec_bam | 1           | 0    |
| 4   | phi_bmq  | 0.752429    | 0    |
| 4   | mer_bmq  | 3           | 0    |
| 4   | mcs_bmq  | 0.833333    | 0    |
| 4   | uwec_bmq | 0.666667    | 0    |
| 4   | phi_boz  | 0.916823    | 0    |
| 4   | mer_boz  | 0           | 0    |
| 4   | mcs_boz  | 0           | 0    |
| 4   | uwec_boz | 1           | 0    |
| 4   | phi_dtm  | 0.809387    | 0    |
| 4   | mer_dtm  | 0.5         | 0    |
| 4   | mcs_dtm  | 0.333333    | 0    |
| 4   | uwec_dtm | 0.666667    | 0    |
| 4   | phi_ful  | 0.980054    | 0    |
| 4   | mer_ful  | 3           | 0    |
| 4   | mcs_ful  | 0.75        | 0    |
| 4   | uwec_ful | 0.625       | 0    |
| 4   | phi_mey  | 0.768726    | 0    |
| 4   | mer_mey  | 0           | 0    |
| 4   | mcs_mey  | 0           | 0    |
| 4   | uwec_mey | 0.5         | 0    |
| 4   | phi_mku  | 0.908601    | 0    |
| 4   | mer_mku  | 12          | 0    |
| 4   | mcs_mku  | 0.923077    | 0    |
| 4   | uwec_mku | 0.538462    | 0    |
| 4   | phi_kao  | 0.970238    | 0    |
| 4   | mer_kao  | 25.3333     | 0    |
| 4   | mcs_kao  | 0.974359    | 0    |
| 4   | uwec_kao | 1           | 0    |
| 4   | phi_myk  | 0.7953      | 0    |
| 4   | mer_myk  | 10          | 0    |
| 4   | mcs_myk  | 0.909091    | 0    |
| 4   | uwec_myk | 0.5         | 0    |
| 4   | phi_ses  | 0.836952    | 0    |
| 4   | mer_ses  | 0           | 0    |
| 4   | mcs_ses  | 0           | 0    |
| 4   | uwec_ses | 1           | 0    |
| 4   | phi_spp  | 0.853018    | 0    |
| 4   | mer_spp  | 6.66667     | 0    |
| 4   | mcs_spp  | 0.909091    | 0    |
| 4   | uwec_spp | 0.545455    | 0    |
| 4   | phi_snk  | 0.781673    | 0    |
| 4   | mer_snk  | 0.333333    | 0    |
| 4   | mcs_snk  | 0.333333    | 0    |
| 4   | uwec_snk | 0.666667    | 0    |
| 4   | phi_taq  | 0.84844     | 0    |
| 4   | mer_taq  | 0.5         | 0    |
| 4   | mcs_taq  | 0.5         | 0    |
| 4   | uwec_taq | 1           | 0    |
| 4   | phi      | 0.849638    | 0    |

| Run | Metric | Final value | Step |
|-----|--------|-------------|------|
| 4   | mer    | 5.16667     | 0    |
| 4   | mcs    | 0.865169    | 0    |
| 4   | uwec   | 0.769663    | 0    |

| Run | Inline claim | Value    | Phase |
|-----|--------------|----------|-------|
| 4   | phi          | 0.849638 | eval  |
| 4   | mer          | 5.16667  | eval  |
| 4   | mcs          | 0.865169 | eval  |
| 4   | uwec         | 0.769663 | eval  |

**r2: DPO arm**

| Run | Metric                   | Final value | Step |
|-----|--------------------------|-------------|------|
| 3   | loss                     | 0.544094    | 500  |
| 3   | rewards_chosen           | 0.191217    | 500  |
| 3   | rewards_rejected         | -0.234148   | 500  |
| 3   | rewards_accuracies       | 0.6875      | 500  |
| 3   | rewards_margins          | 0.425366    | 500  |
| 3   | eval_rewards_chosen      | -0.00811578 | 500  |
| 3   | eval_rewards_rejected    | -0.419341   | 500  |
| 3   | eval_rewards_accuracies  | 0.719444    | 500  |
| 3   | eval_rewards_margins     | 0.411225    | 500  |
| 3   | eval_loss                | 0.54551     | 500  |
| 3   | eval_runtime             | 77.3164     | 500  |
| 3   | eval_samples_per_second  | 4.656       | 500  |
| 3   | eval_steps_per_second    | 0.582       | 500  |
| 3   | eval_entropy             | 0.887941    | 500  |
| 3   | eval_num_tokens          | 3.0435e+07  | 500  |
| 3   | eval_logits_chosen       | 10.1072     | 500  |
| 3   | eval_logits_rejected     | 10.1393     | 500  |
| 3   | eval_mean_token_accuracy | 0.850163    | 500  |
| 3   | eval_logps_chosen        | -1112.08    | 500  |
| 3   | eval_logps_rejected      | -1114.43    | 500  |
| 3   | epoch                    | 2.4642      | 500  |
| 3   | phi                      | 0.847792    | 500  |
| 3   | phi_bam                  | 0.947662    | 500  |
| 3   | phi_bmq                  | 0.788389    | 500  |
| 3   | phi_mku                  | 0.874554    | 500  |
| 3   | phi_myk                  | 0.858518    | 500  |
| 3   | phi_ses                  | 0.673515    | 500  |
| 3   | phi_dtm                  | 0.875799    | 500  |
| 3   | phi_snk                  | 0.839542    | 500  |
| 3   | phi_boz                  | 0.880553    | 500  |
| 3   | phi_ful                  | 0.872129    | 500  |
| 3   | phi_spp                  | 0.999123    | 500  |
| 3   | phi_taq                  | 0.715931    | 500  |
| 4   | phi_bam                  | 0.823658    | 0    |
| 4   | mer_bam                  | 0           | 0    |
| 4   | mcs_bam                  | 0           | 0    |
| 4   | uwec_bam                 | 1           | 0    |
| 4   | phi_bmq                  | 0.752429    | 0    |

| Run | Metric   | Final value | Step |
|-----|----------|-------------|------|
| 4   | mer_bmq  | 3           | 0    |
| 4   | mcs_bmq  | 0.833333    | 0    |
| 4   | uwec_bmq | 0.666667    | 0    |
| 4   | phi_boz  | 0.916823    | 0    |
| 4   | mer_boz  | 0           | 0    |
| 4   | mcs_boz  | 0           | 0    |
| 4   | uwec_boz | 1           | 0    |
| 4   | phi_dtm  | 0.809387    | 0    |
| 4   | mer_dtm  | 0.5         | 0    |
| 4   | mcs_dtm  | 0.333333    | 0    |
| 4   | uwec_dtm | 0.666667    | 0    |
| 4   | phi_ful  | 0.980054    | 0    |
| 4   | mer_ful  | 3           | 0    |
| 4   | mcs_ful  | 0.75        | 0    |
| 4   | uwec_ful | 0.625       | 0    |
| 4   | phi_mey  | 0.768726    | 0    |
| 4   | mer_mey  | 0           | 0    |
| 4   | mcs_mey  | 0           | 0    |
| 4   | uwec_mey | 0.5         | 0    |
| 4   | phi_mku  | 0.908601    | 0    |
| 4   | mer_mku  | 12          | 0    |
| 4   | mcs_mku  | 0.923077    | 0    |
| 4   | uwec_mku | 0.538462    | 0    |
| 4   | phi_kao  | 0.970238    | 0    |
| 4   | mer_kao  | 25.3333     | 0    |
| 4   | mcs_kao  | 0.974359    | 0    |
| 4   | uwec_kao | 1           | 0    |
| 4   | phi_myk  | 0.7953      | 0    |
| 4   | mer_myk  | 10          | 0    |
| 4   | mcs_myk  | 0.909091    | 0    |
| 4   | uwec_myk | 0.5         | 0    |
| 4   | phi_ses  | 0.836952    | 0    |
| 4   | mer_ses  | 0           | 0    |
| 4   | mcs_ses  | 0           | 0    |
| 4   | uwec_ses | 1           | 0    |
| 4   | phi_spp  | 0.853018    | 0    |
| 4   | mer_spp  | 6.66667     | 0    |
| 4   | mcs_spp  | 0.909091    | 0    |
| 4   | uwec_spp | 0.545455    | 0    |
| 4   | phi_snk  | 0.781673    | 0    |
| 4   | mer_snk  | 0.333333    | 0    |
| 4   | mcs_snk  | 0.333333    | 0    |
| 4   | uwec_snk | 0.666667    | 0    |
| 4   | phi_taq  | 0.84844     | 0    |
| 4   | mer_taq  | 0.5         | 0    |
| 4   | mcs_taq  | 0.5         | 0    |
| 4   | uwec_taq | 1           | 0    |
| 4   | phi      | 0.849638    | 0    |
| 4   | mer      | 5.16667     | 0    |
| 4   | mcs      | 0.865169    | 0    |

| Run | Metric | Final value | Step |
|-----|--------|-------------|------|
| 4   | uwec   | 0.769663    | 0    |

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| Run | Inline claim | Value    | Phase |
|-----|--------------|----------|-------|
| 4   | phi          | 0.849638 | eval  |
| 4   | mer          | 5.16667  | eval  |
| 4   | mcs          | 0.865169 | eval  |
| 4   | uwec         | 0.769663 | eval  |

**r3: APO arm**

| Run | Metric                   | Final value | Step |
|-----|--------------------------|-------------|------|
| 3   | loss                     | 0.899064    | 500  |
| 3   | rewards_chosen           | 0.491311    | 500  |
| 3   | rewards_rejected         | 0.0567837   | 500  |
| 3   | rewards_accuracies       | 0.8125      | 500  |
| 3   | rewards_margins          | 0.434527    | 500  |
| 3   | eval_rewards_chosen      | 0.295494    | 500  |
| 3   | eval_rewards_rejected    | -0.0857438  | 500  |
| 3   | eval_rewards_accuracies  | 0.669444    | 500  |
| 3   | eval_rewards_margins     | 0.381238    | 500  |
| 3   | eval_loss                | 0.911244    | 500  |
| 3   | eval_runtime             | 77.3757     | 500  |
| 3   | eval_samples_per_second  | 4.653       | 500  |
| 3   | eval_steps_per_second    | 0.582       | 500  |
| 3   | eval_entropy             | 0.887046    | 500  |
| 3   | eval_num_tokens          | 3.0435e+07  | 500  |
| 3   | eval_logits_chosen       | 10.1392     | 500  |
| 3   | eval_logits_rejected     | 10.1657     | 500  |
| 3   | eval_mean_token_accuracy | 0.85068     | 500  |
| 3   | eval_logps_chosen        | -1109.39    | 500  |
| 3   | eval_logps_rejected      | -1111.44    | 500  |
| 3   | epoch                    | 2.4642      | 500  |
| 3   | phi                      | 0.847792    | 500  |
| 3   | phi_bam                  | 0.947662    | 500  |
| 3   | phi_bmq                  | 0.788389    | 500  |
| 3   | phi_mku                  | 0.874554    | 500  |
| 3   | phi_myk                  | 0.858518    | 500  |
| 3   | phi_ses                  | 0.673515    | 500  |
| 3   | phi_dtm                  | 0.875799    | 500  |
| 3   | phi_snk                  | 0.839542    | 500  |
| 3   | phi_boz                  | 0.880553    | 500  |
| 3   | phi_ful                  | 0.872129    | 500  |
| 3   | phi_spp                  | 0.999123    | 500  |
| 3   | phi_taq                  | 0.715931    | 500  |
| 4   | phi_bam                  | 0.823658    | 0    |
| 4   | mer_bam                  | 0           | 0    |
| 4   | mcs_bam                  | 0           | 0    |
| 4   | uwec_bam                 | 1           | 0    |
| 4   | phi_bmq                  | 0.752429    | 0    |
| 4   | mer_bmq                  | 3           | 0    |
| 4   | mcs_bmq                  | 0.833333    | 0    |

| Run | Metric   | Final value | Step |
|-----|----------|-------------|------|
| 4   | uwec_bmq | 0.666667    | 0    |
| 4   | phi_boz  | 0.916823    | 0    |
| 4   | mer_boz  | 0           | 0    |
| 4   | mcs_boz  | 0           | 0    |
| 4   | uwec_boz | 1           | 0    |
| 4   | phi_dtm  | 0.809387    | 0    |
| 4   | mer_dtm  | 0.5         | 0    |
| 4   | mcs_dtm  | 0.333333    | 0    |
| 4   | uwec_dtm | 0.666667    | 0    |
| 4   | phi_ful  | 0.980054    | 0    |
| 4   | mer_ful  | 3           | 0    |
| 4   | mcs_ful  | 0.75        | 0    |
| 4   | uwec_ful | 0.625       | 0    |
| 4   | phi_mey  | 0.768726    | 0    |
| 4   | mer_mey  | 0           | 0    |
| 4   | mcs_mey  | 0           | 0    |
| 4   | uwec_mey | 0.5         | 0    |
| 4   | phi_mku  | 0.908601    | 0    |
| 4   | mer_mku  | 12          | 0    |
| 4   | mcs_mku  | 0.923077    | 0    |
| 4   | uwec_mku | 0.538462    | 0    |
| 4   | phi_kao  | 0.970238    | 0    |
| 4   | mer_kao  | 25.3333     | 0    |
| 4   | mcs_kao  | 0.974359    | 0    |
| 4   | uwec_kao | 1           | 0    |
| 4   | phi_myk  | 0.7953      | 0    |
| 4   | mer_myk  | 10          | 0    |
| 4   | mcs_myk  | 0.909091    | 0    |
| 4   | uwec_myk | 0.5         | 0    |
| 4   | phi_ses  | 0.836952    | 0    |
| 4   | mer_ses  | 0           | 0    |
| 4   | mcs_ses  | 0           | 0    |
| 4   | uwec_ses | 1           | 0    |
| 4   | phi_spp  | 0.853018    | 0    |
| 4   | mer_spp  | 6.66667     | 0    |
| 4   | mcs_spp  | 0.909091    | 0    |
| 4   | uwec_spp | 0.545455    | 0    |
| 4   | phi_snk  | 0.781673    | 0    |
| 4   | mer_snk  | 0.333333    | 0    |
| 4   | mcs_snk  | 0.333333    | 0    |
| 4   | uwec_snk | 0.666667    | 0    |
| 4   | phi_taq  | 0.84844     | 0    |
| 4   | mer_taq  | 0.5         | 0    |
| 4   | mcs_taq  | 0.5         | 0    |
| 4   | uwec_taq | 1           | 0    |
| 4   | phi      | 0.849638    | 0    |
| 4   | mer      | 5.16667     | 0    |
| 4   | mcs      | 0.865169    | 0    |
| 4   | uwec     | 0.769663    | 0    |

| Run | Inline claim | Value    | Phase |
|-----|--------------|----------|-------|
| 4   | phi          | 0.849638 | eval  |
| 4   | mer          | 5.16667  | eval  |
| 4   | mcs          | 0.865169 | eval  |
| 4   | uwec         | 0.769663 | eval  |

### 3 Setup

| Report | Item             | Value                                           |
|--------|------------------|-------------------------------------------------|
| r1     | Operating system | linux / amd64                                   |
| r1     | CPU              | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores |
| r1     | GPU              | NVIDIA GeForce RTX 5060 Ti, 16311 MiB           |
| r1     | Memory           | 31.0 GB                                         |
| r1     | Total run time   | 2189.8 s                                        |
| r2     | Operating system | linux / amd64                                   |
| r2     | CPU              | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores |
| r2     | GPU              | NVIDIA GeForce RTX 5060 Ti, 16311 MiB           |
| r2     | Memory           | 31.0 GB                                         |
| r2     | Total run time   | 4033.2 s                                        |
| r3     | Operating system | linux / amd64                                   |
| r3     | CPU              | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores |
| r3     | GPU              | NVIDIA GeForce RTX 5060 Ti, 16311 MiB           |
| r3     | Memory           | 31.0 GB                                         |
| r3     | Total run time   | 3993.8 s                                        |

### 4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

| File            | SHA-256                                                           | Size   |
|-----------------|-------------------------------------------------------------------|--------|
| r1.pdf          | 4a6ef8c7ed1f6bcd3c1af4057fd1870532ebb5d4c95cbc7b7fec9701f1458b97  | 8.8 MB |
| r1.kvbundle.zip | 199eb8b7c51638eae96fa0c03f3f5834f1b16c35debd164a5a4999b69388553a  | 345 KB |
| r2.pdf          | d6cd2489aa7e223b79fe5ede12864e2a689c1da03093eb7bcf2ed0a25bc35454  | 9.9 MB |
| r2.kvbundle.zip | b88486ea00ad32dd4baaed944b0a415e59f7166d1664ca1c60cbe4f3263f521c  | 346 KB |
| r3.pdf          | 442c651ab9985d1b4821bcabdaa2bca5a57c6bf0b7ce707ef98cacf1649a4c7a0 | 9.8 MB |
| r3.kvbundle.zip | 62f040330bb576322043e72745134b995ff5b696bdfa1f8c62fa006027af6947  | 348 KB |

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Listed, not endorsed. Review checks the artifact, never the science. Records are never edited; corrections are new versions.

## References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.
- [2] Sealed experiment report r1 (SFT arm), kv:2610.00001v1/r1. Data hash 1c1699a5bb61643823134d4b20b8cbf76989b4df4327e1bd9ed91816f2e2d177. <https://kveritas.org/records/2610.00001v1/r1.pdf>
- [3] Sealed experiment report r2 (DPO arm), kv:2610.00001v1/r2. Data hash b2c776c815223b6b9677d4aa01595ece5ca4c79c02fde0bbdd361ba783e7b4d8. <https://kveritas.org/records/2610.00001v1/r2.pdf>
- [4] Sealed experiment report r3 (APO arm), kv:2610.00001v1/r3. Data hash 3d35b38ed6fb3a13c0ac03d020b070894bb4f1d5469f322ef456aca5b4eff9a6. <https://kveritas.org/records/2610.00001v1/r3.pdf>